

Wafer Map Failure Pattern Classification

Foundations of Deep Learning Project



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From Wafer Maps to Defect Classification

What is a Wafer Map?

A wafer map is a **2D categorical representation** of functional and defective dies across a semiconductor wafer.

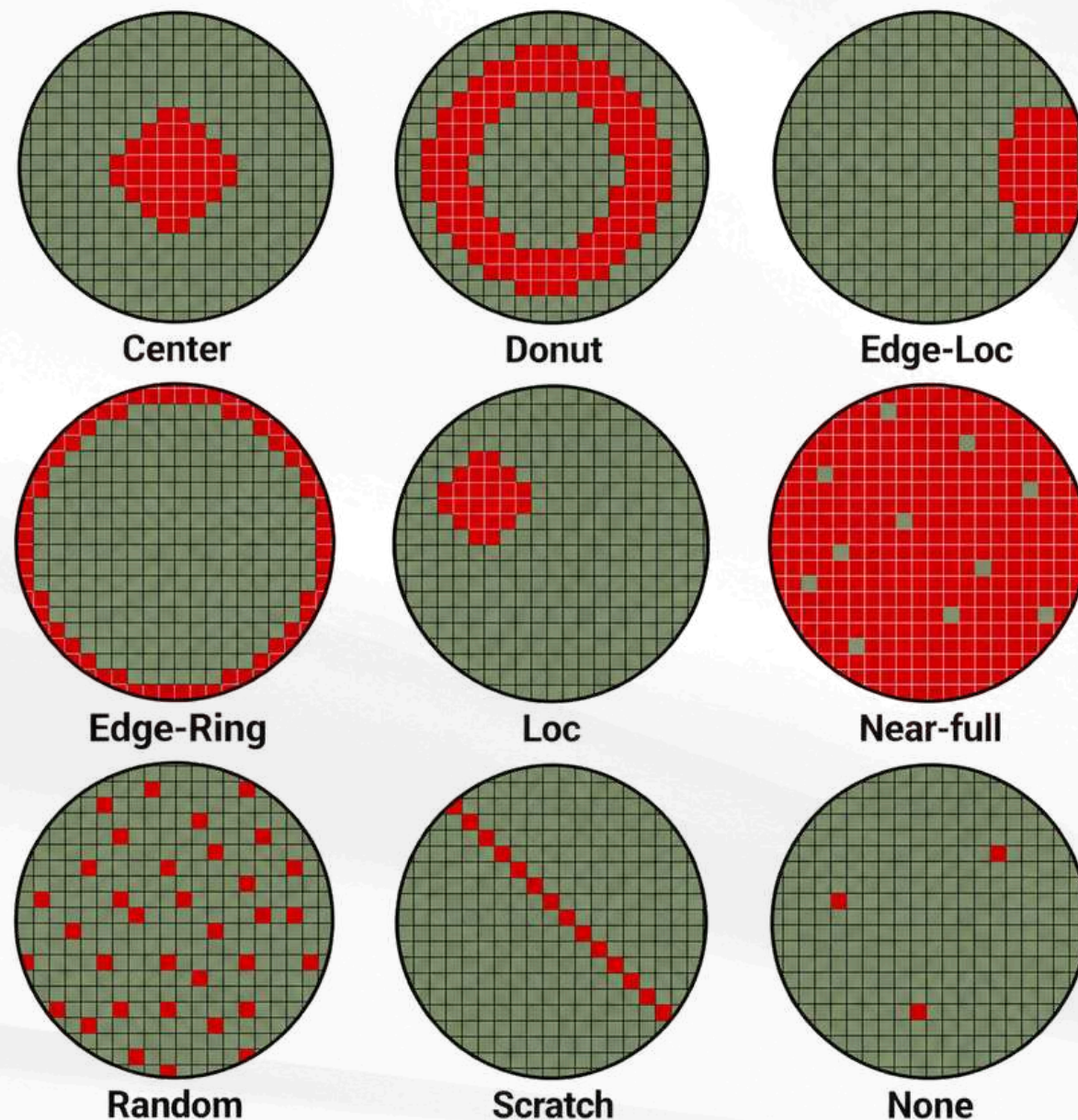
WM-811K Dataset

811,457 rows and 6 columns: *dieSize*, *failureType*, *lotName*, *trainTestLabel*, *waferIndex* and *waferMap* :

- **172,950 labeled maps** (21.3%) used for supervised learning
- **638,507 unlabeled maps** (78.7%) reserved for possible future semi-supervised learning

Goal

Classify each labelled wafer map into one of nine failure-pattern classes.



Exploratory Analysis and Dataset Split

Map sizes

- ▶ 632 different shapes
- ▶ Median size: 36 × 35
- ▶ Range: 6–300 × 3–205

Lot Name and Failure Patterns

Wafer maps from the same production lot are not independent and frequently share similar failure patterns.



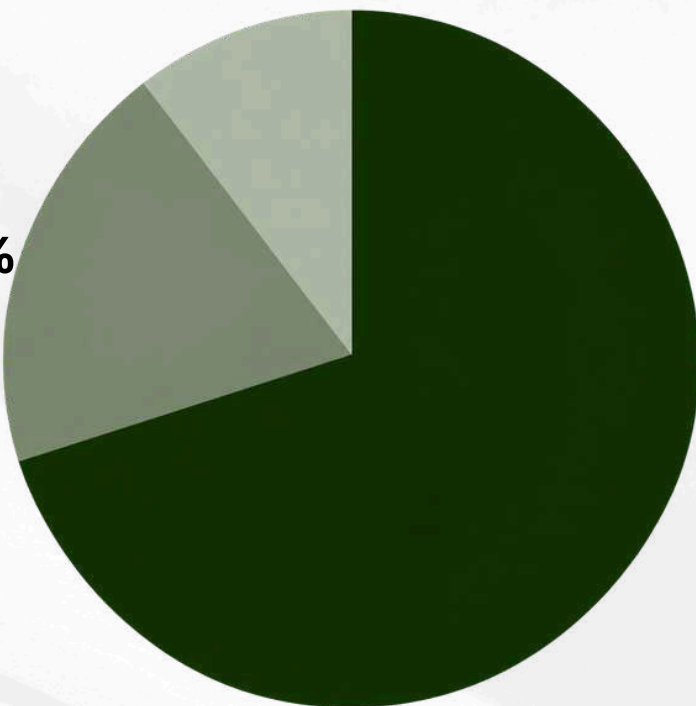
GROUP-AWARE SPLIT

Whole lots were assigned to one subset while preserving class proportions.

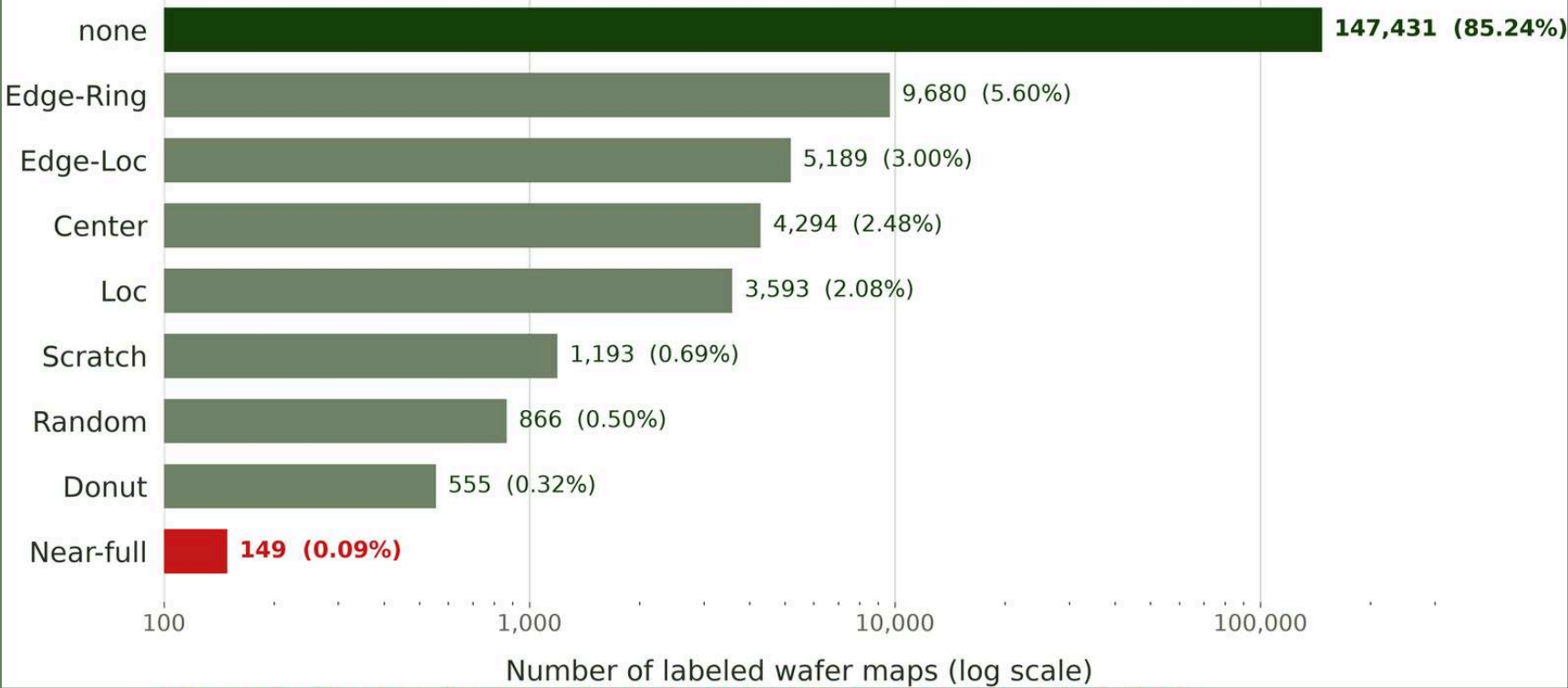
Test Set: 10%

Validation Set: 20%

Training Set: 70%



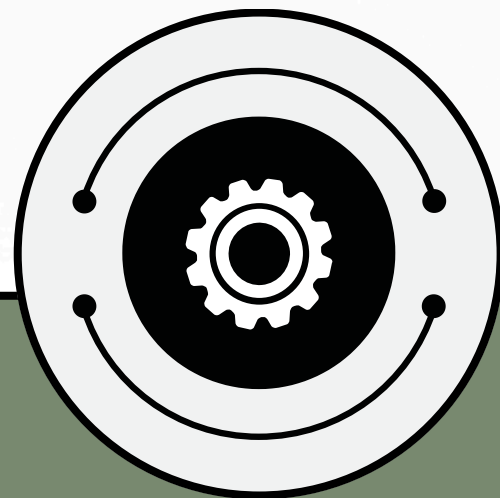
The labeled dataset is strongly imbalanced



Evaluation Protocol

1

Per-class precision, recall and F1



TRAIN: 70%
(121,063 wafers)

Fit model parameters
using the training split
only

2

*Primary metric: **macro-F1***
weights all nine classes
equally

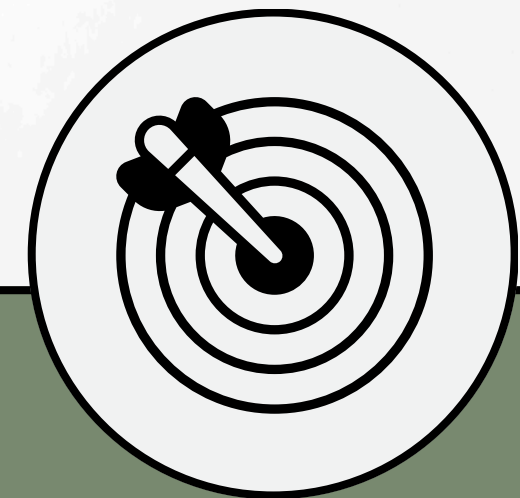


VALIDATION: 20%
(34,591 wafers)

1. Early stopping
2. Checkpoint selection
3. Model and
hyperparameter selection

3

Validation and test
retain the natural class
distribution



TEST: 10%
(17,296 wafers)

One final evaluation after
all modeling decisions are
fixed

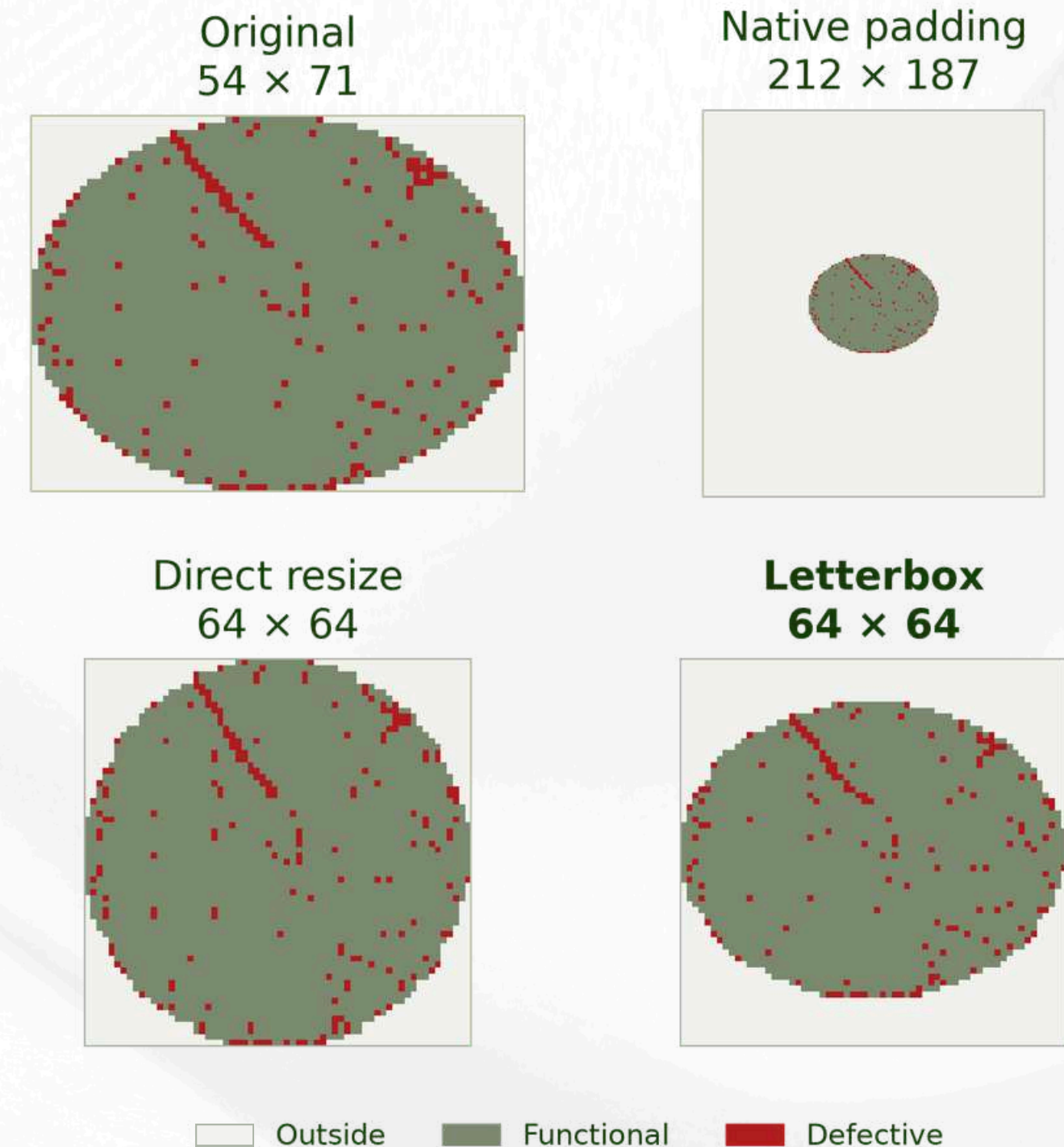
Image Preprocessing

We compared four pipelines on 900 training maps (100 per class).

Letterboxing preserves wafer proportions at 64×64 or 128×128 .

Nearest-neighbour resizing preserves categorical states.

Three one-hot channels encode **outside wafer**, **functional die** and **defective die**.



Baseline CNN

A cheap baseline CNN was used to test training choices one factor at a time and traced *experiments* on W&B's platform

CNN's Architecture

Compact reference · ~157k parameters

INPUT

64 × 64 wafer map · 3 one-hot channels



FEATURE EXTRACTOR

Conv + BN + ReLU · channels 16 → 32 → 64



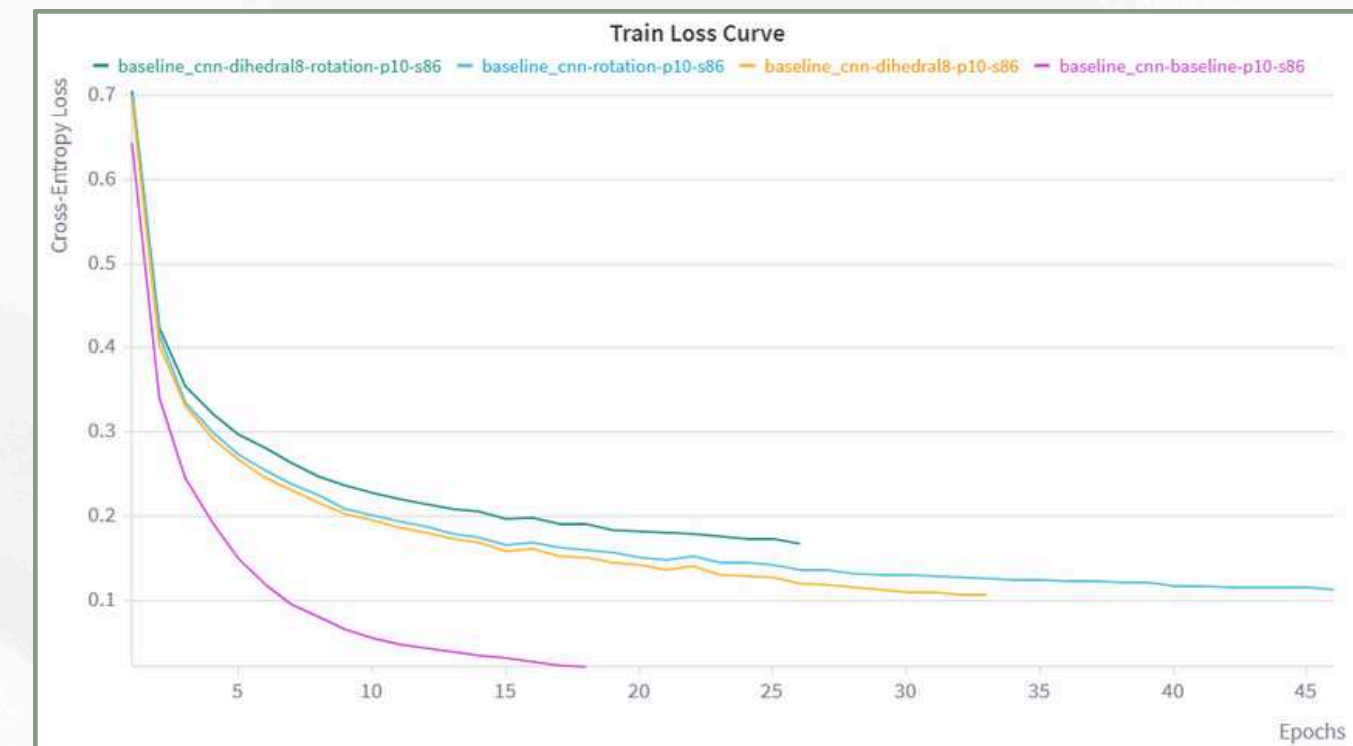
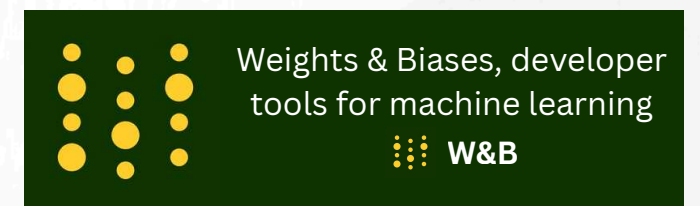
CLASSIFIER

4 × 4 pooling → Dense 128 → 9 classes

Experiments explored

- Class imbalance handling
- Data augmentation
- Input encoding and resolution

These experiments guided the training setup for subsequent models.



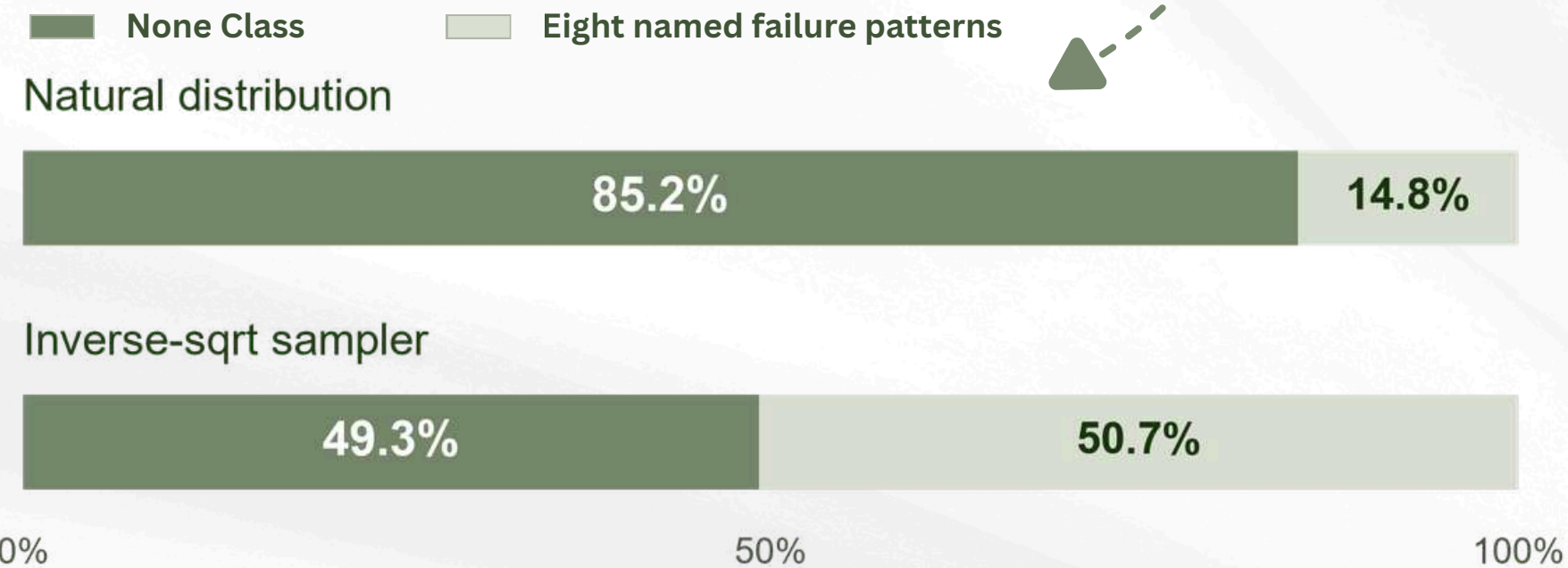
Training dynamics tracked in W&B

Class Imbalance

Five strategies were considered:

- Standard cross-entropy
- Effective-number weighting
- Inverse-sqrt weighting
- Focal loss
- **Inverse-sqrt sampling**

Expected class share during training



Scratch: 0.69% → 4.43% Near-full: 0.09% → 1.56%

121,063 draws per epoch · training split only

Initial four-epoch screening:

Validation macro-F1 · mean ± SD, 3 seeds



Based on the initial screening results, we chose inverse-sqrt sampling with standard cross-entropy.

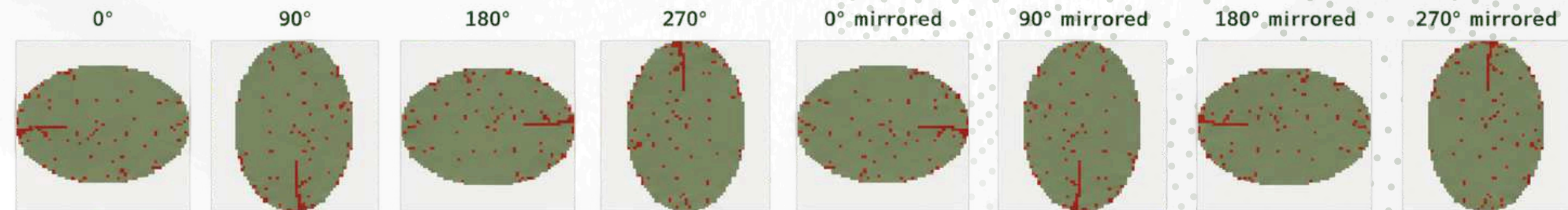
*Weighting each map by $\frac{1}{\sqrt{n_c}}$ makes rare patterns come up **more often, not equally often.***

Data Augmentations

- D4 symmetries — eight exact views. Nothing is resampled.— **Competitive performance.**
- Free-angle rotations — any angle, nearest neighbour, with $p = 0.5$ — **Performed best on the experiment.**

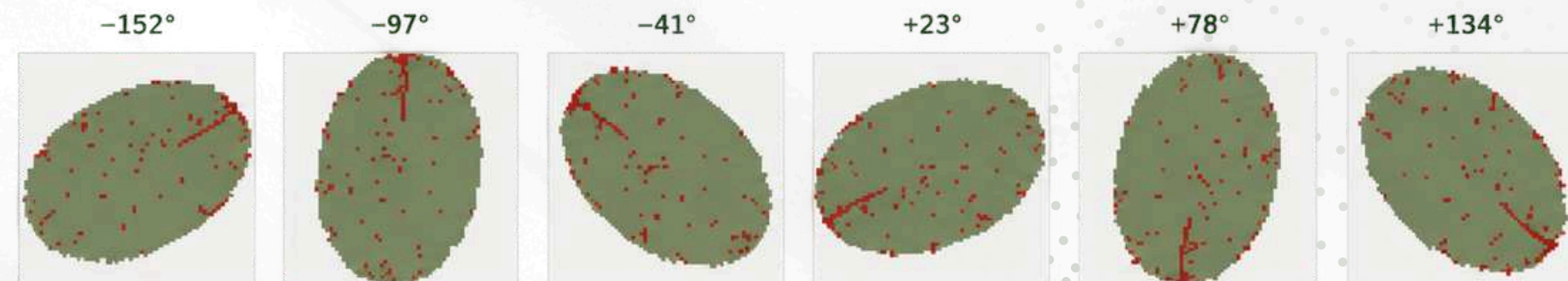
D4 symmetries

eight exact views — every cell is moved, none invented



Free-angle rotation

any angle, resampled — nearest neighbour keeps the three states



□ outside wafer
■ functional die
■ defective die

Baseline CNN, validation macro-F1

none: 0.8173 | D4: 0.8587 | rotation: 0.8800 | D4 + rotation: 0.8677

Eight exact views beat none. Random angle views beat eight.

Dilation Impact

WHY DILATION?

Capture global wafer patterns while preserving thin defects.

Dilated convolutions expand the receptive field without extra kernel weights or downsampling.

CONTROLLED COMPARISON

Training the same models, changing the dilation rates.

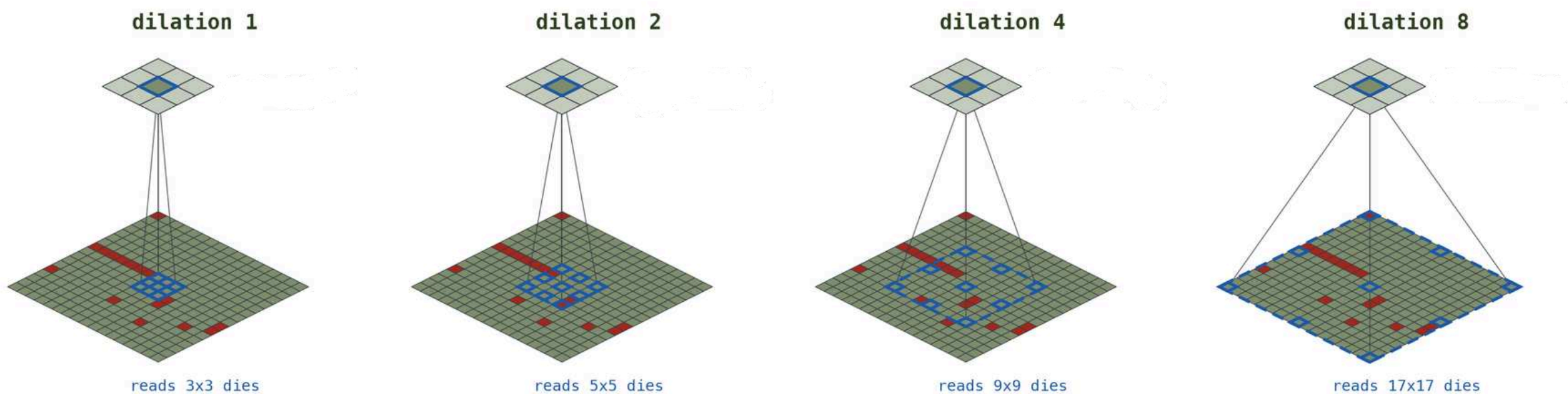
Compact CNN – 298k parameters

Receptive field: 19x19 → 63x63

Rates: all 1 → (1, 2, 4, 8)x2

Validation Macro-F1

MODEL	CONTROL	DILATED
Compact CNN	0.8217	0.8852
ConvNeXt	0.8988	0.8957



Same nine weights and the same cost in all four. Only the reach changes and nothing downsamples.

functional die

defective die

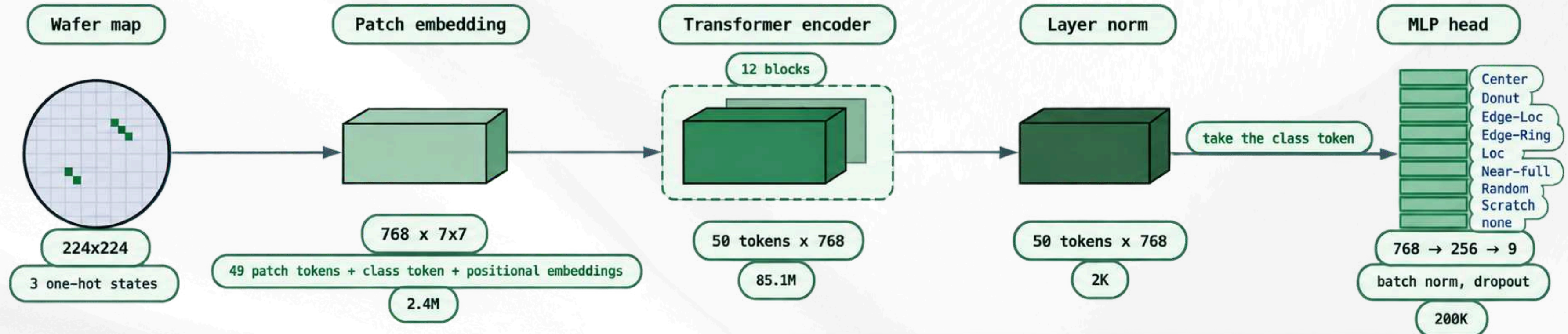
kernel tap

Selected Models

Model:	Parameters:	Reason:
ViT-B/32, fine-tuned	88M	The best validation with TTA and class thresholds
ResNet-34, fine-tuned	21M	The best on plain validation
Our ConvNeXt, from scratch	2.7M	The best model we built, comparable metric, while being smaller size
Our dilated CNN, from scratch	0.3M	Highlights dilation effect, ~300x fewer parameters than ViT-B/32

Fine-tuned ViT-B/32

Pretrained Tranformer | 87.7M parameters | 224x224 input

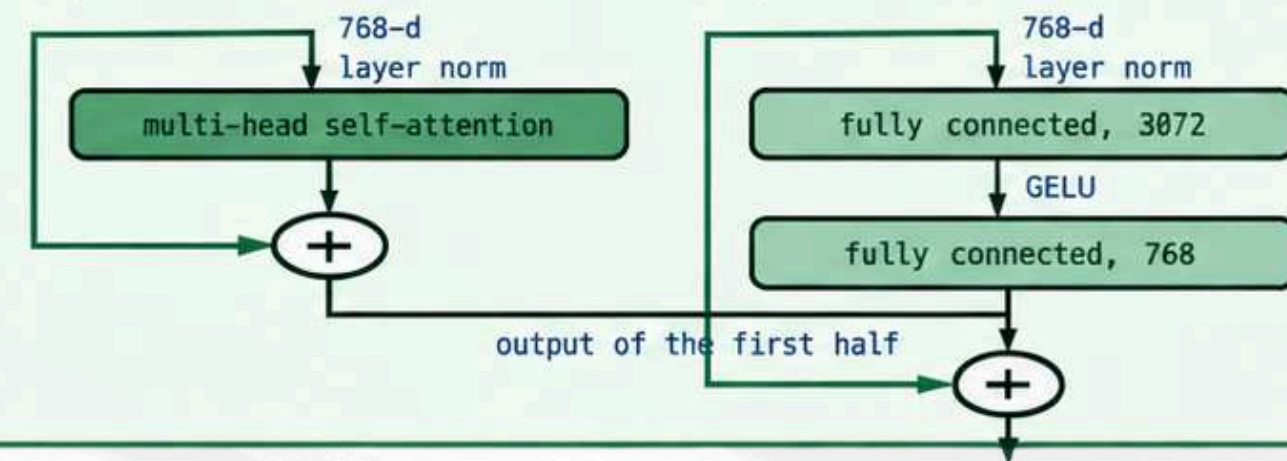


Why this architecture

Attention is global from the first block, so an Edge-Ring or a Donut is one relation between distant patches rather than something a stack of local kernels has to grow into. The cost is resolution, since a one-die-wide Scratch is a fraction of one 32x32 patch, and ImageNet supplies priors that 172,950 labelled wafers cannot.

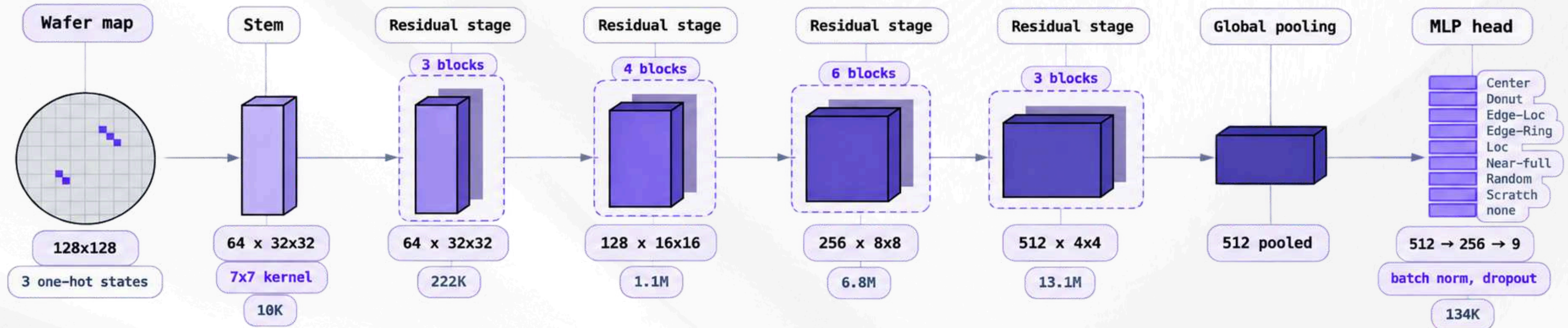
Inside one transformer block

each of the 50 tokens forms a query, a key and a value and attends to all 50, itself included, through 12 heads of 64 features. The two halves are sequential, not parallel: the second reads what the first produced



Fine-tuned ResNet-34

Best on the Held-Out split | 21.4M parameters | 128x128 input

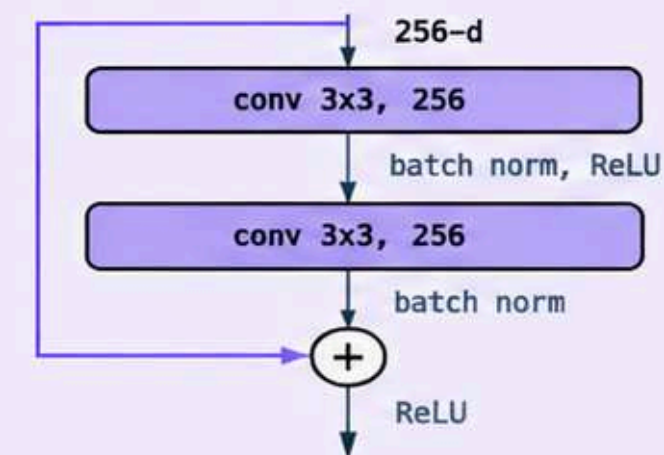


Why this architecture

The plainest thing that could work, and the control everything else is read against. ImageNet transfers here because its early layers detect edges and blobs, which is most of what a defect pattern is. Depth stays affordable because the stem drops resolution fourfold before the first block runs.

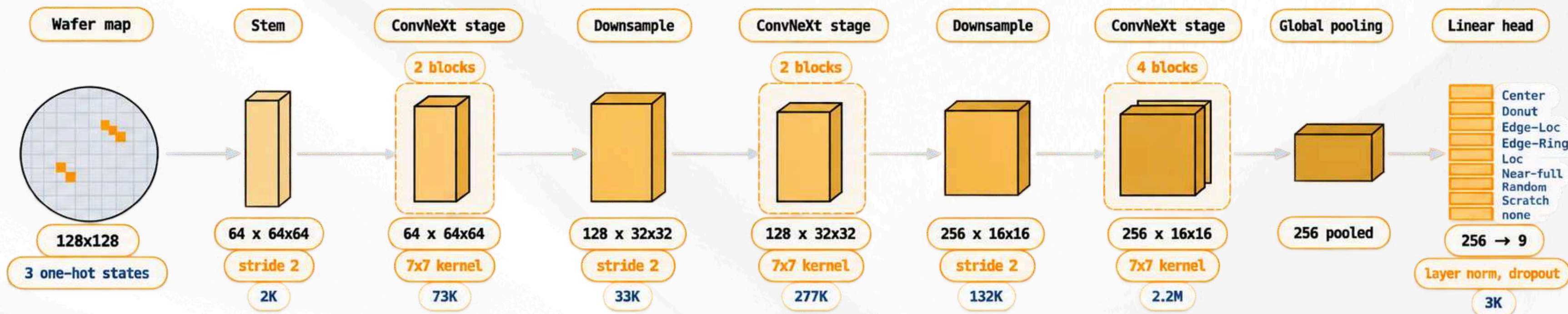
Inside one residual block

two 3x3 convolutions with a residual connection; the first block of each new stage strides by 2 and takes a 1x1 projection shortcut, since the identity no longer matches



Our ConvNeXt from scratch

Best model from scratch | 2.7M parameters | 128x128 input

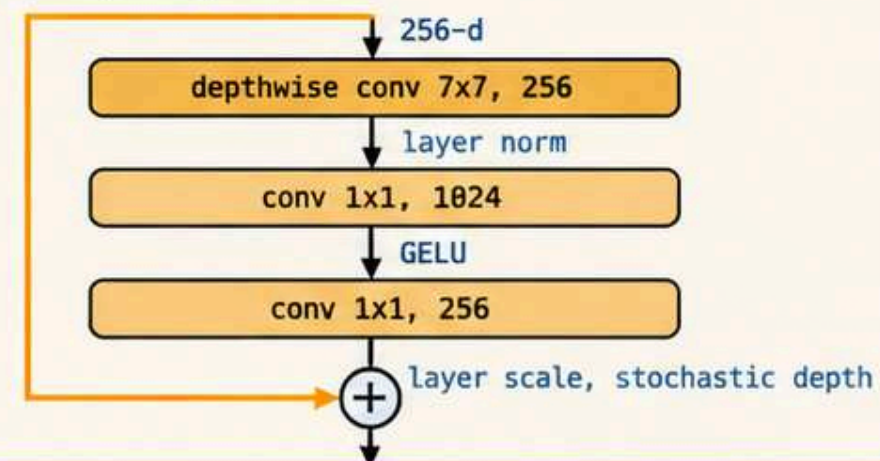


Why this architecture

Keeps the convolution and adopts everything else a transformer does: large depthwise kernels, an inverted bottleneck, LayerNorm and GELU, with far fewer activations than a ResNet. It isolates how much of a transformer's advantage is attention and how much is the surrounding design. The stem is stride 2, not the reference 4x4 patchify, which would cut the spatial grid by 94% immediately and risk diluting very narrow Scratch defects.

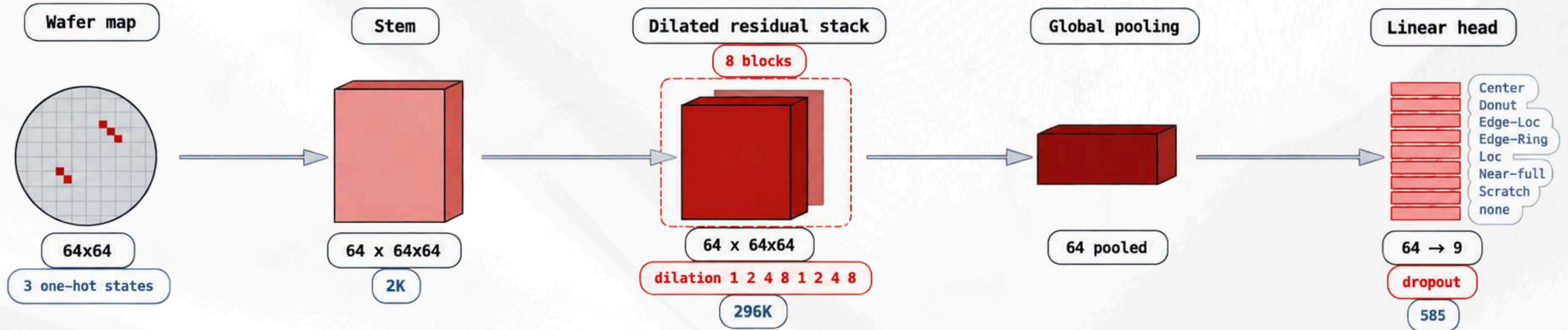
Inside one ConvNeXt block

one normalisation and one activation for the whole block, and the 7x7 mixing is depthwise, which is what makes it affordable. Shown at the widest stage; the same block runs at every width, always $C \rightarrow 4C \rightarrow C$.



Our Dilated CNN from scratch

298K parameters | 64x64 input

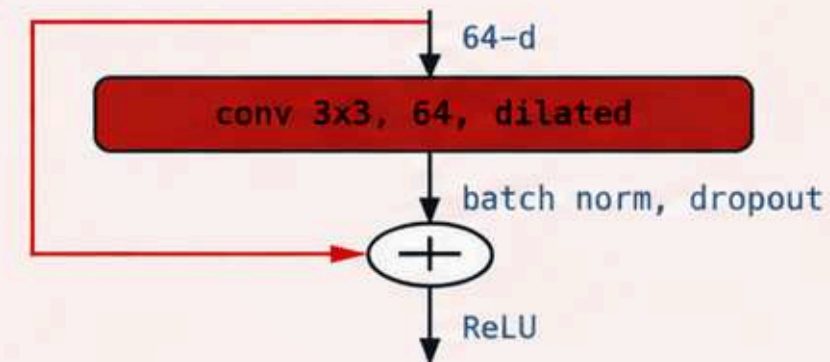


Why this architecture

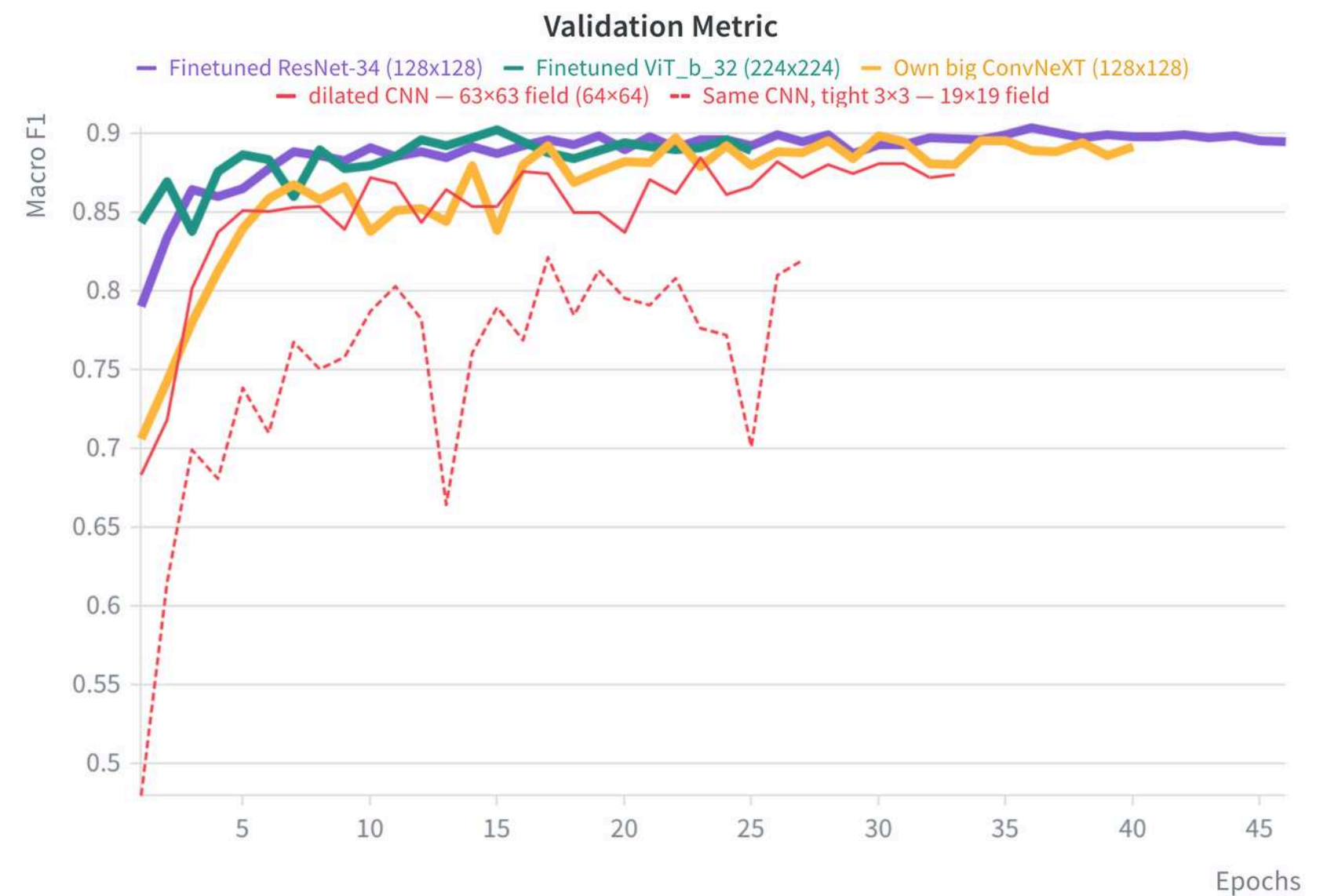
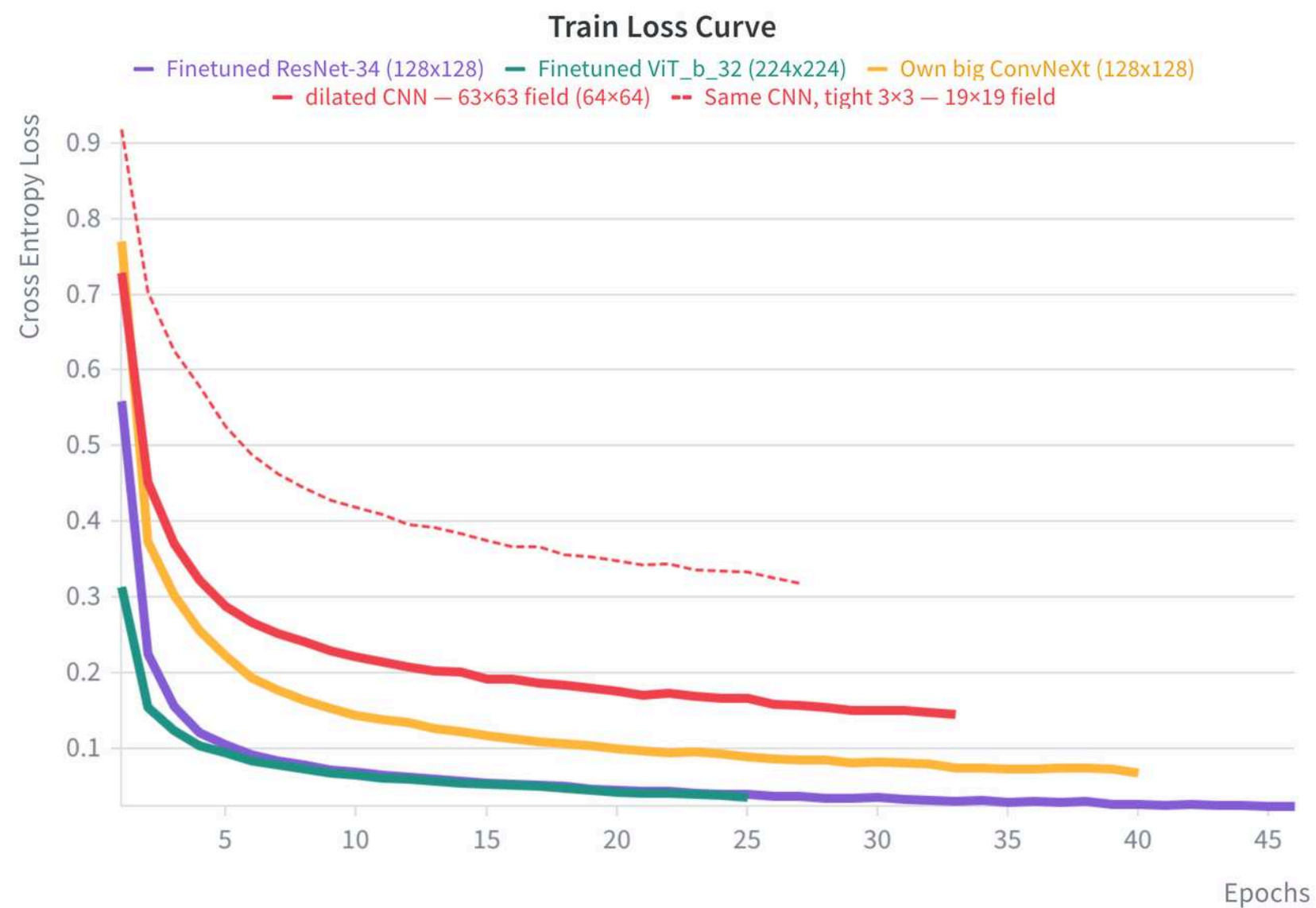
No downsampling preserves one-die-wide defects throughout the feature maps, and the receptive field comes from dilation rather than pooling: 298K parameters reach 63x63 on a 64x64 map. Nothing is pooled until the very end.

Inside one dilated block

one dilated 3x3 convolution per residual block, with the rate cycling 1, 2, 4, 8, 1, 2, 4, 8 across the stack. Cycling widens the receptive field while mitigating the gridding a single repeated rate causes; nothing downsamples.



Training and Validation Results



- Fine-tuned ResNet-34 and ViT-B/32 achieved similar reported scores.
- Our ConvNeXt remained competitive with far fewer parameters.
- Dilation markedly improves the CNN compared to the control.

Validation & Holdout

PREDICTION REFINEMENT

We used **Test-time Augmentation**: average predictions across eight rotated and mirrored views of each wafer

Class-score weighting: we tuned one multiplier per class (w_c) on validation to maximise macro-F1, then weighted the predicted probabilities before selecting the final class.



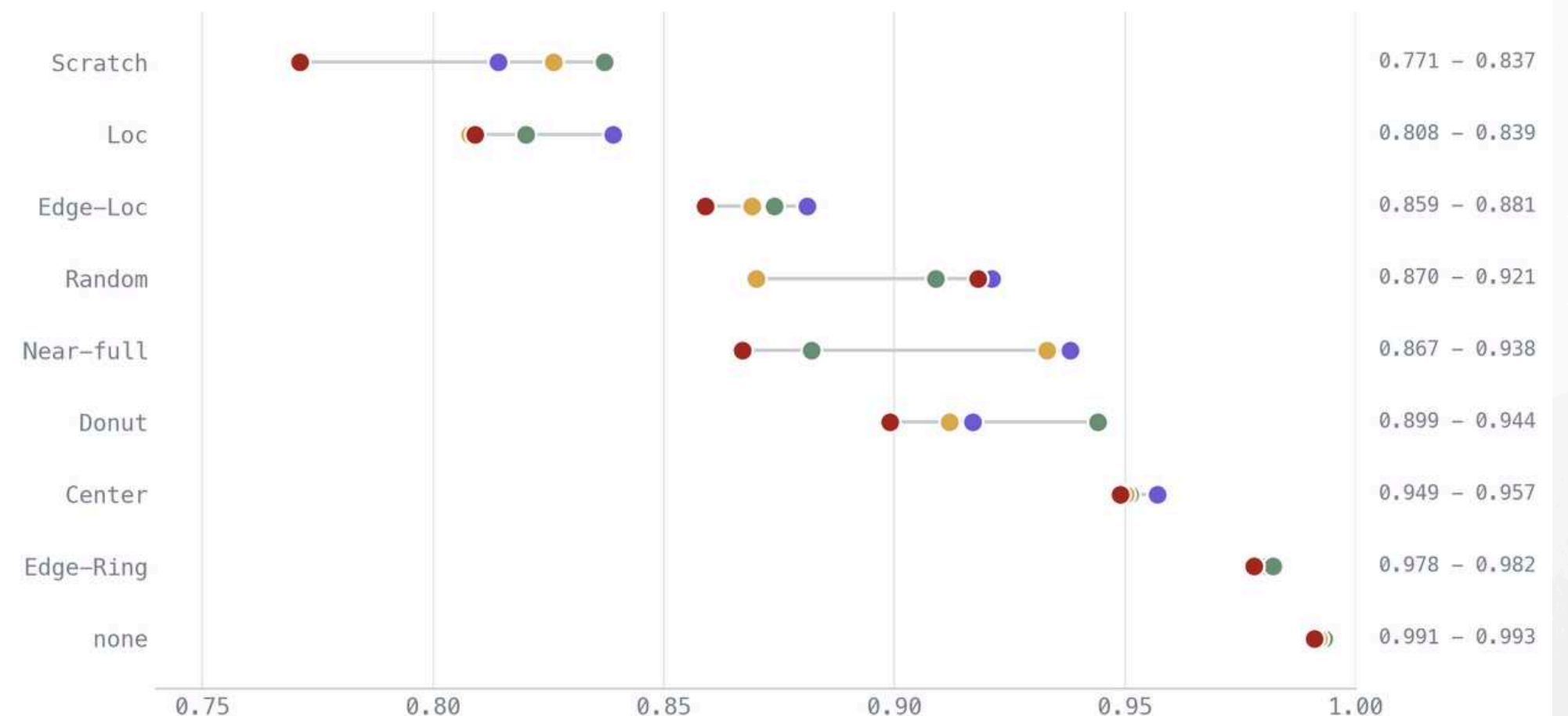
KEY RESULTS

- TTA and class-score weighting **improved all models** without retraining.
- Validation and holdout scores remained **close**.
- Center, Edge-Ring and none showed consistently **high F1** across models.

Macro-F1

	validation		held-out test	
	plain	tuned	plain	tuned
ResNet-34, fine-tuned	0.9042	0.9148	0.9016	0.9156
ViT-B/32, fine-tuned	0.9029	0.9160	0.9004	0.9103
Our ConvNeXt	0.8988	0.9109	0.8977	0.9042
Our dilated CNN	0.8852	0.9005	0.8858	0.8934

Per-class F1, held-out test, tuned

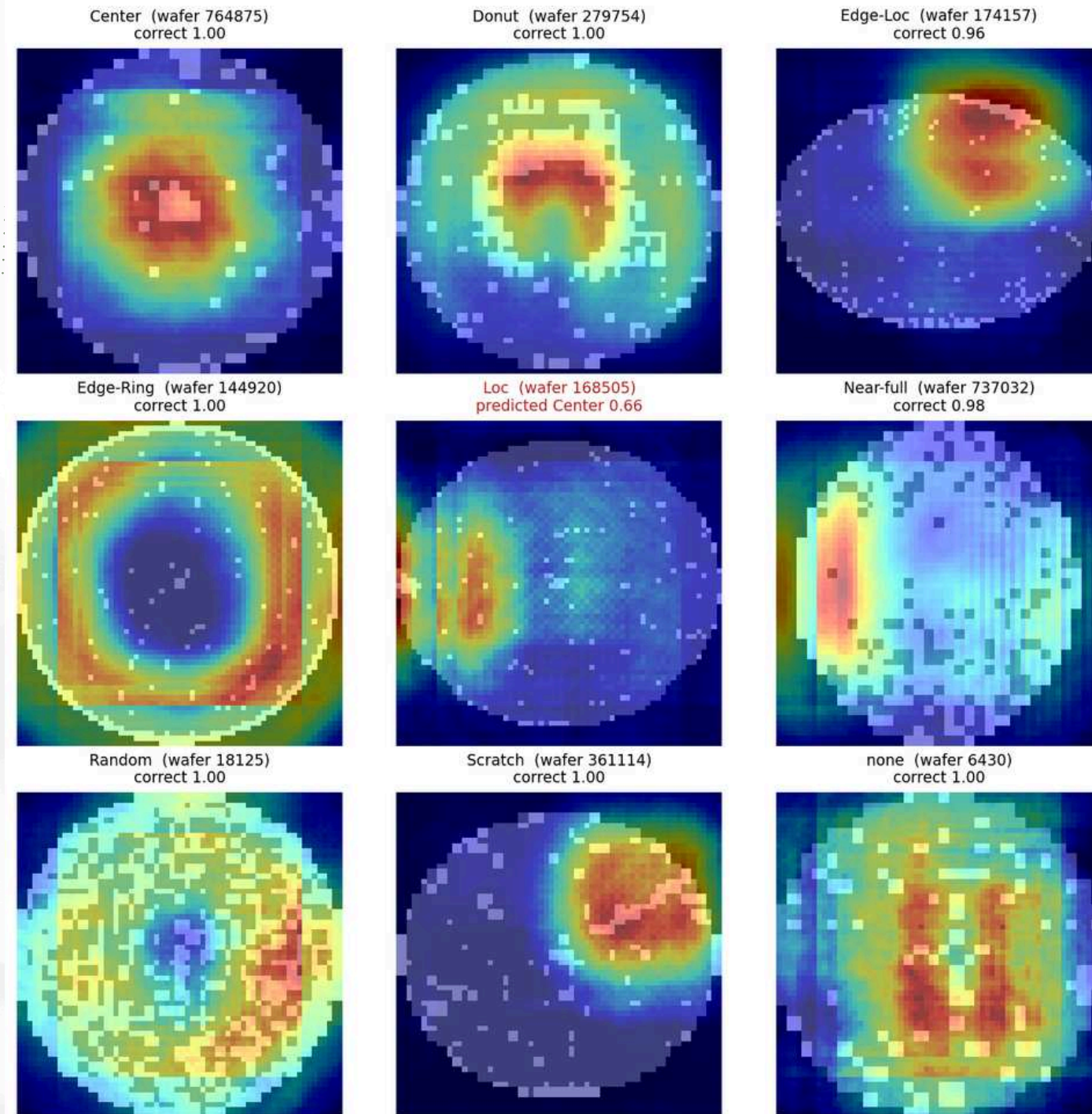


Qualitative Visual Analysis

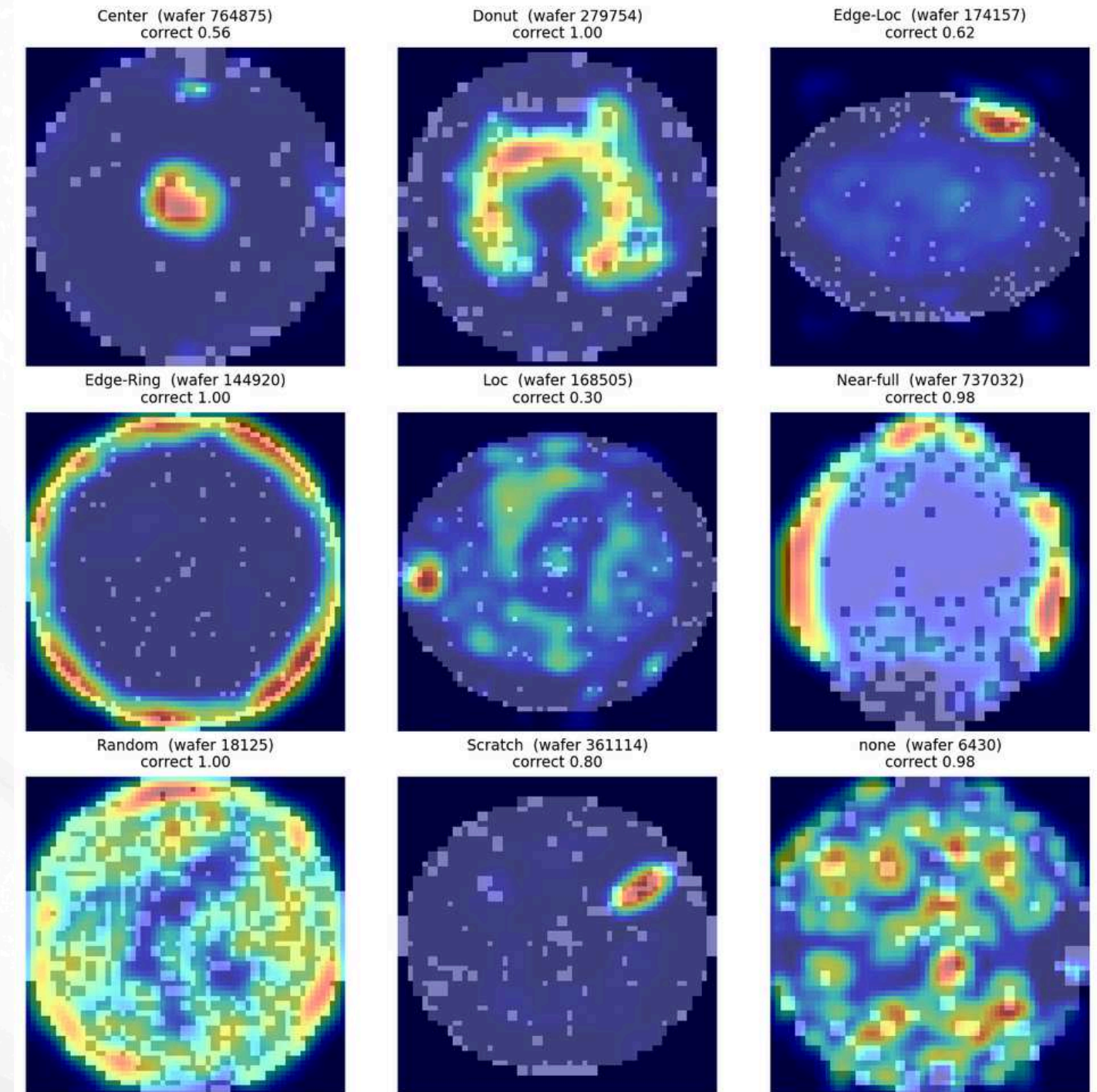
Grad-CAM: Dilated CNN vs Normal CNN

How do the maps change when modifying the dilation within the same architecture?

Grad-CAM: v31-dilated_rotation_64



Grad-CAM: v31-dilated_control_64

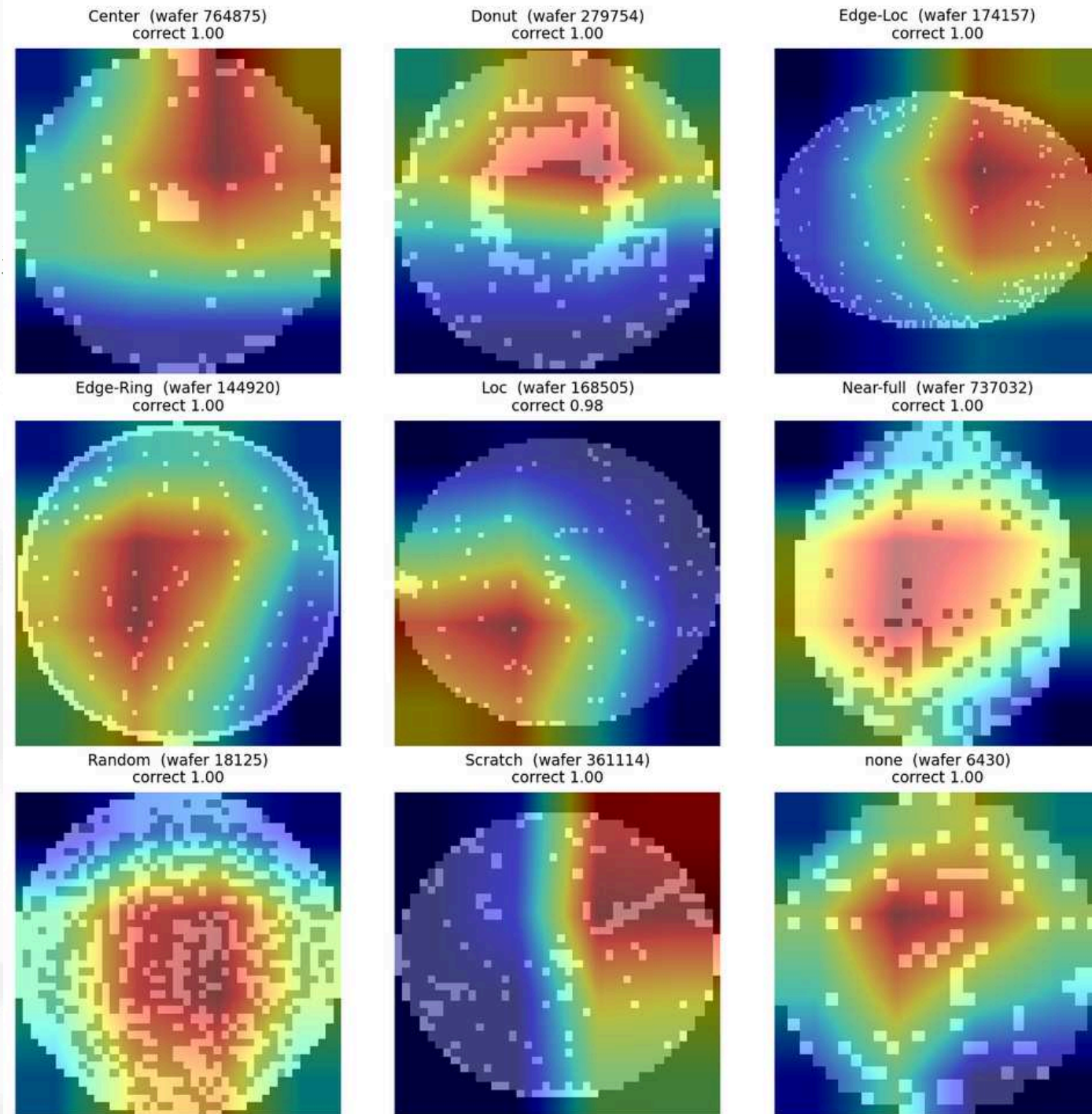


Qualitative Visual Analysis

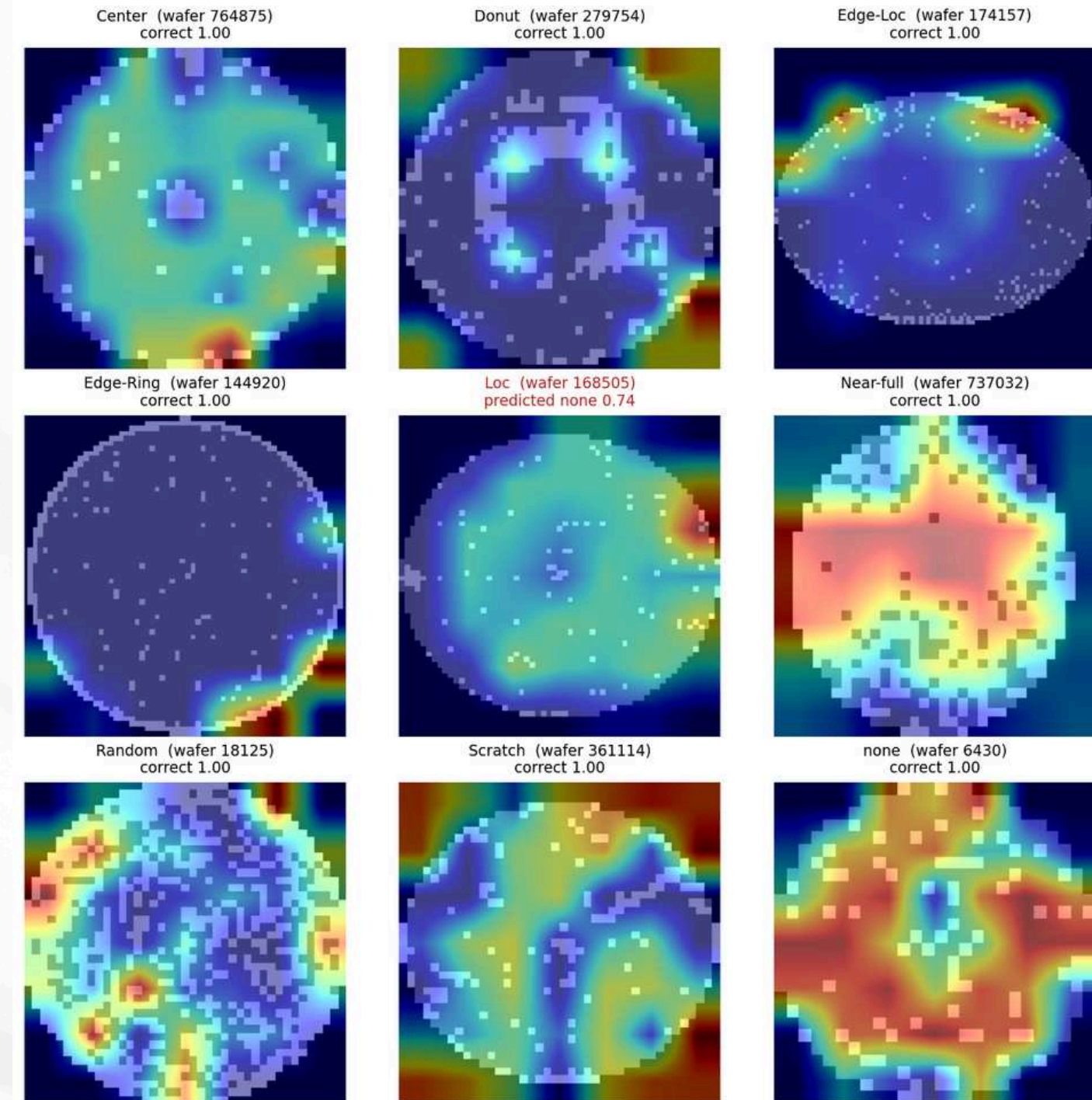
Top Finetuned Models: ResNet vs ViT

What visual differences emerge between two fine-tuned models from different families?

Grad-CAM: v32-resnet34_finetune



Grad-CAM: v32-vit_b_32_finetune

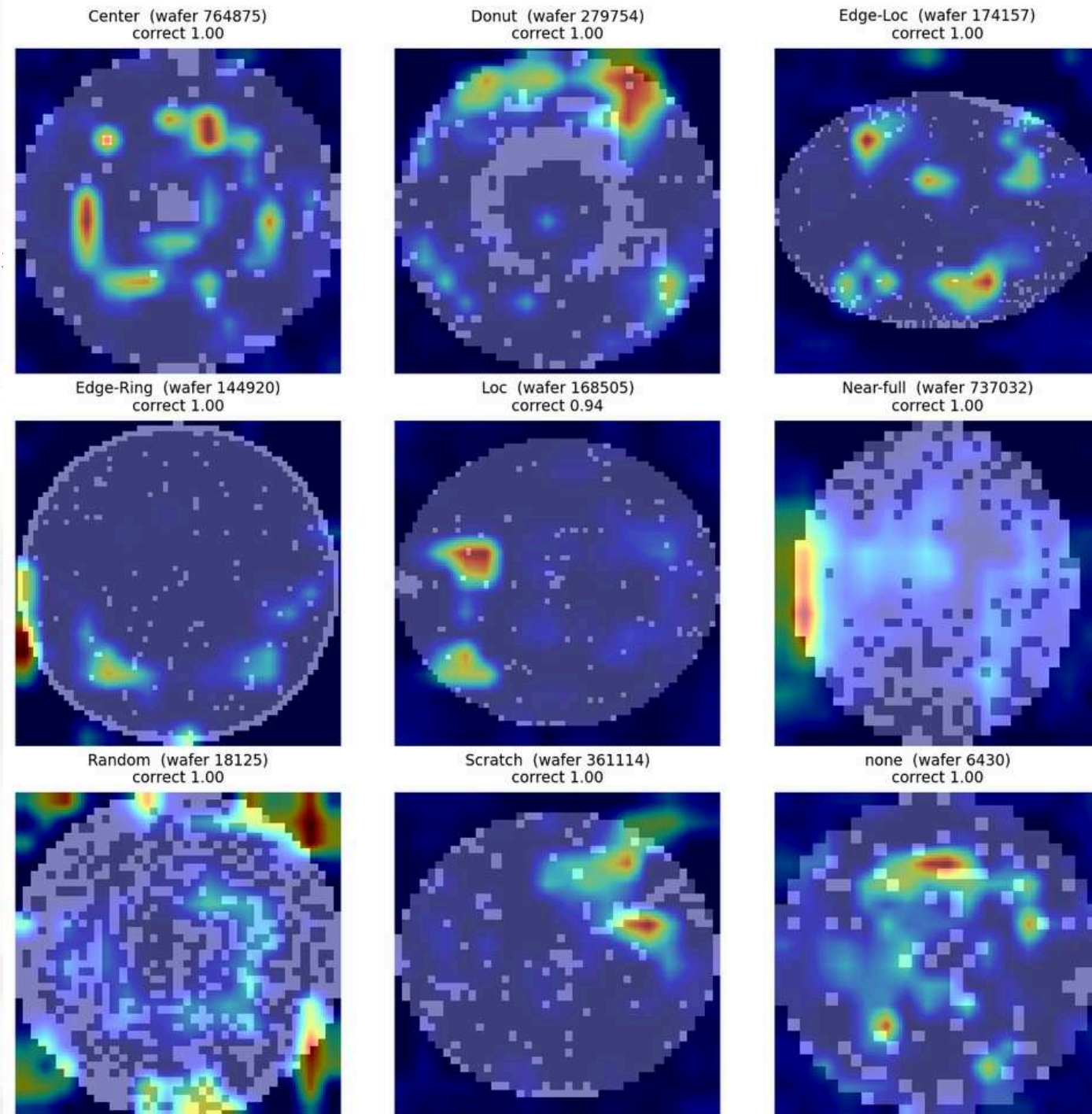


Qualitative Visual Analysis

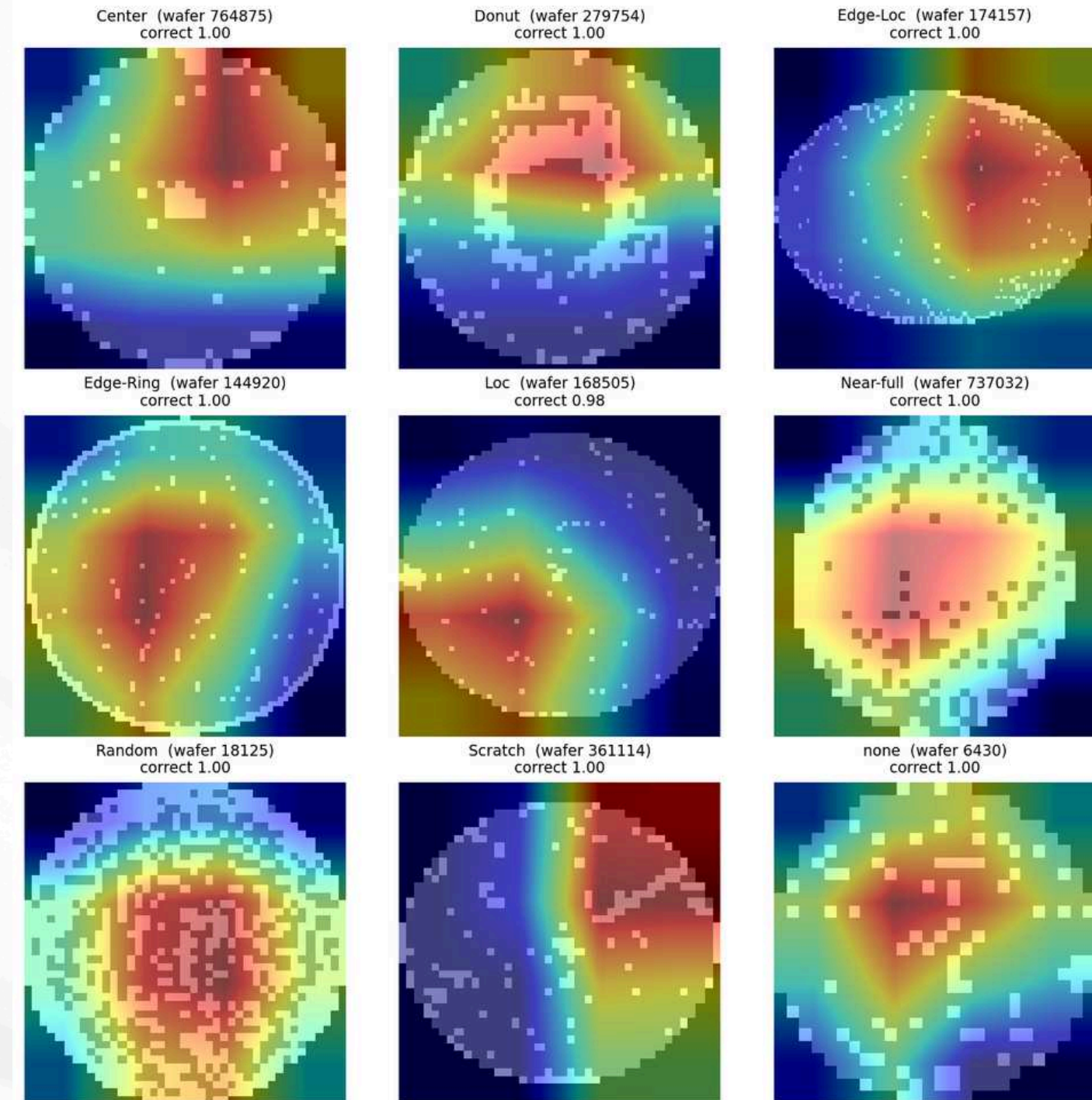
Our model from scratch (ConvNext) vs. Best model (ResNet-34)

How do the maps from our Scratch model differ from those of the model with the best reported test macro-F1?

Grad-CAM: v28-convnext_big_128



Grad-CAM: v32-resnet34_finetune



Failed Experiments

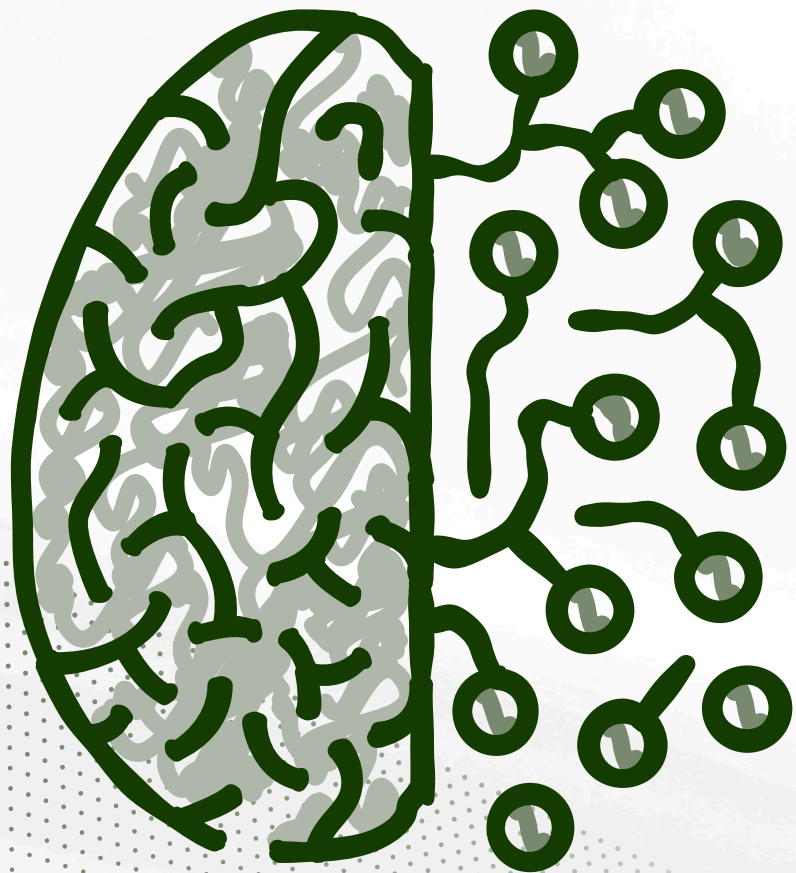
- **Imbalance:** longer runs didn't show a clear gain from inverse sqrt sampling or alternative losses like focal loss, compared to the initial experiment.
- **Spatial attention:** the CBAM layer didn't improve the baseline CNN model; thus it wasn't considered for more models.
- **Capacity:** increasing our ConvNeXt from 414k to 2.68M parameters at 64×64 changed validation macro-F1 by -0.0021 ; more parameters did not guarantee better results.
- **Transfer learning:** no frozen encoder achieved satisfactory results. ResNet18 improved from 0.7351 (frozen) to 0.8938 (fine-tuned).
- **Dilation:** helped the full-resolution CNN, but gave no clear gain on ConvNeXt.

Limitations & Next Steps

- Improved validation schema with GroupKFold is a better approach than a train/val/test split, but requires more resources.
- Repeat leading configurations across seeds: single run differences can be unstable.
- Experiment with semi-supervised learning using unlabeled maps
- Try using generative models like GANs, VAEs, diffusion models, to upsample underrepresented classes.



Conclusions



- 1 Compact **models** trained from **scratch** remain **competitive** with much larger pretrained networks.
- 2 Controlled baseline **experiments** guided training choices: random-angle rotations improved validation performance, while **fine-tuning** outperformed frozen features.
- 3 **Dilation** improved the compact full-resolution CNN **without adding parameters**, but its benefit depended on the architecture.
- 4 **Test-time augmentation** and validation-tuned **class-score weighting** improved all four models on the held-out test, without retraining.
- 5 **Model selection** requires balancing predictive performance, uncertainty and resource requirements, rather than relying on a single score.

**THANKS FOR
YOUR ATTENTION!**